

WAYNE COUNTY AP, KY, USA

WMO: 720379

Lat: 36.855N

Lon: 84.856W

Elev: 294

StdP: 97.85

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 63882

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-9.9	-18.3	0.8	-10.1	-15.1	1.0	-7.6	11.0	8.6	9.8	9.5	1.9	10	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	33.8	24.0	32.6	24.0	31.4	23.7	26.2	30.8	25.4	30.1	24.7	29.4	2.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.7	29.2	24.1	19.7	28.0	23.3	18.8	27.0	82.7	30.7	79.2	30.2	76.2	29.3	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-19.0	27.4	4.7	1.5	-22.4	28.4	-25.1	29.3	-27.8	30.2	-31.2	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	1.4	3.4	8.7	14.0	19.0	23.1	24.3	23.8	20.7	14.1	7.6	4.4	(6)
(7)		DBStd	9.45	6.99	6.36	6.07	5.10	4.27	2.67	2.74	2.64	3.56	5.01	5.30	5.42	(7)
(8)		HDD10.0	912	279	199	99	20	1	0	0	0	0	18	108	188	(8)
(9)		HDD18.3	2363	525	418	301	145	48	2	1	1	18	149	325	431	(9)
(10)		CDD10.0	2284	12	15	58	141	279	393	444	427	320	144	34	16	(10)
(11)		CDD18.3	693	0	0	2	16	67	145	187	169	88	18	2	0	(11)
(12)		CDH23.3	6811	0	1	33	203	623	1404	1763	1636	915	224	8	0	(12)
(13)	CDH26.7	2624	0	0	3	32	176	547	759	697	353	57	1	0	(13)	
(14)	Wind	WSAvg	2.8	3.4	3.6	3.6	3.5	2.9	2.3	1.8	1.6	2.2	2.6	2.9	3.4	(14)
(15)	Precipitation	PrecAvg	1315	129	98	127	110	127	119	120	103	84	73	90	137	(15)
(16)		PrecMax	1609	207	192	275	233	238	241	191	204	161	131	175	363	(16)
(17)		PrecMin	1016	60	37	53	31	52	50	57	33	7	18	17	60	(17)
(18)		PrecStd	167	34	40	55	46	48	50	40	45	48	32	36	68	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.5	22.0	26.8	29.0	31.9	34.1	36.9	36.0	34.8	31.4	24.6	20.3	(19)
(20)			MCWB	15.2	15.8	18.1	18.5	22.9	23.1	24.6	23.9	22.5	21.7	17.2	15.9	(20)
(21)		2%	DB	17.5	19.3	23.9	27.5	30.1	32.6	33.4	33.6	32.2	28.0	21.3	18.2	(21)
(22)			MCWB	14.0	14.2	16.4	18.2	21.7	23.7	24.8	24.2	22.7	19.7	15.3	14.6	(22)
(23)		5%	DB	15.5	16.8	21.3	25.9	28.7	31.4	32.2	32.1	30.7	26.0	19.1	15.8	(23)
(24)			MCWB	12.6	12.8	14.7	17.2	21.2	23.5	24.7	24.0	22.3	18.4	14.0	12.8	(24)
(25)		10%	DB	12.9	14.5	19.0	23.9	27.2	30.0	30.9	30.6	28.6	23.4	17.2	14.0	(25)
(26)			MCWB	10.3	10.8	13.2	16.7	20.6	22.9	24.2	23.6	21.0	17.6	13.0	11.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.8	17.1	18.7	21.2	24.3	26.0	27.5	27.3	25.0	23.4	19.0	17.3	(27)
(28)			MCD B	18.6	20.4	24.7	27.4	30.0	31.0	31.9	32.6	30.2	28.8	22.6	19.8	(28)
(29)		2%	WB	14.7	15.6	17.4	19.9	22.9	25.2	26.5	25.8	24.0	21.6	17.4	15.1	(29)
(30)			MCD B	16.9	18.7	22.8	25.3	28.2	30.3	31.0	30.6	29.9	25.6	19.7	17.1	(30)
(31)		5%	WB	12.9	13.6	16.0	18.6	22.2	24.4	25.7	25.1	23.2	19.9	15.4	13.6	(31)
(32)			MCD B	15.2	15.8	20.0	23.4	27.3	29.5	30.2	29.9	28.6	23.9	17.7	15.3	(32)
(33)		10%	WB	10.6	11.4	14.5	17.5	21.3	23.7	25.0	24.3	22.2	18.5	13.5	11.6	(33)
(34)			MCD B	12.4	13.8	18.0	22.1	25.9	28.6	29.3	28.8	26.7	22.3	16.5	13.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.9	11.6	12.8	14.3	12.2	12.6	12.0	12.8	13.2	14.1	13.5	11.2	(35)
(36)			MCD BR	12.3	14.7	15.8	17.2	14.0	14.1	14.1	14.6	15.9	16.8	15.8	13.3	(36)
(37)			MCWBR	9.5	10.4	9.5	9.0	6.6	6.2	6.2	6.3	7.0	8.8	10.0	10.3	(37)
(38)		5% WB	MCD BR	10.7	12.0	13.6	13.4	12.5	12.0	11.3	12.0	13.1	12.7	11.0	11.8	(38)
(39)			MCWBR	8.9	9.8	8.8	7.7	6.2	5.8	5.9	6.0	6.2	7.6	8.4	9.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.308	0.318	0.351	0.394	0.445	0.465	0.499	0.503	0.413	0.362	0.324	0.305	(40)	
(41)		taud	2.499	2.460	2.411	2.284	2.176	2.152	2.081	2.056	2.289	2.425	2.497	2.517	(41)	
(42)		Ebn at Noon	887	920	915	887	844	824	792	779	837	855	859	866	(42)	
(43)		Edh at Noon	81	94	109	131	148	152	162	162	122	97	80	75	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.07	2.68	3.80	5.02	5.59	6.15	5.78	5.45	4.55	3.52	2.51	1.80	(44)	
(45)		RadStd	0.22	0.37	0.35	0.37	0.49	0.35	0.31	0.28	0.47	0.45	0.27	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	+87

Nomenclature: See separate page